Targeted Maximum Likelihood Estimation (TMLE) for the Average Treatment Effect (ATE)

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Overview

Targeted Maximum Likelihood Estimation (**TMLE**) is a general framework for estimating causal effects that combines ideas from outcome regression (like g-computation) and propensity score methods (like IPTW) to produce estimators that are: - **Doubly robust**: consistent if either the outcome model or the propensity model is correctly specified. - **Efficient**: achieves the lowest possible asymptotic variance when both models are correct. - **Bounded**: respects natural bounds (e.g., probabilities stay in [0,1] for binary outcomes). - Provides valid standard errors and confidence intervals based on the efficient influence function (EIF). - Can flexibly incorporate machine learning methods for nuisance function estimation. - Extensible to complex settings, including longitudinal treatments and time-varying confounding.

R Packages

```
pacman::p_load(
  tmle, # TMLE for point treatment
  ltmle, # Longitudinal TMLE
  SuperLearner # Super Learner for flexible nuisance estimation
)
```

What TMLE is (in One paragraph)

TMLE is a **plug-in estimator** that combines:

- an **outcome regression** like **g-computation** (model $Q(a,w) = \mathbb{E}[Y \mid A = a, W = w]$), and

- a **propensity model** like **IPTW** (model $g(w) = \Pr(A = 1 \mid W = w)$),

and then performs a **targeted update** of the outcome regression using the propensity information so that the estimate is:

- **doubly robust** (consistent if **either** Q or g is correct),
- **efficient** under mild conditions (achieves the semiparametric efficiency bound when both are correct),
- **bounded** (for binary Y, predictions stay in $[0, 1]$),
- and has valid, influence-function–based **standard errors** (no reliance on parametric outcome models).

Intuitively: TMLE starts with a sensible prediction model for Y, then makes a **surgical, one-parameter tweak** guided by the **propensity** to remove residual bias *exactly* in the direction that matters for your estimand.

The Setup and the Target

- Observed data: $O = (W, A, Y)$ with covariates W, treatment $A \in \{0, 1\}$, outcome Y.
- Estimand (ATE):

$$\psi_0 \;=\; \mathbb{E}[Y^1 - Y^0] \;=\; \mathbb{E}[Q_0(1, W) - Q_0(0, W)].$$

- Nuisance functions:
  - Outcome regression $Q_0(a, w) = \mathbb{E}[Y \mid A = a, W = w]$.
  - Propensity $g_0(w) = \Pr(A = 1 \mid W = w)$.

Assumptions (same trio as IPTW/g-comp): **consistency**, **positivity**, and **(sequential) exchangeability**.

The Efficient Influence Function (EIF) and why it Matters

For the ATE, the EIF is

$$D^*(O; \psi, Q, g) \;=\; H(A, W)\,\{Y - Q(A, W)\} \;+\; (Q(1, W) - Q(0, W)) - \psi,$$

where the **clever covariate**

$$H(A, W) \;=\; \frac{A}{g(W)} - \frac{1 - A}{1 - g(W)}.$$

- If your estimator $\hat{\psi}$ is good, the **empirical mean** of $D^*(O; \hat{\psi}, \hat{Q}, \hat{g})$ should be ~ 0.
- TMLE chooses a **tiny update** of Q so that this mean is (approximately) 0—i.e., it solves the **estimating equation** implied by the EIF. That's the "targeting."

The TMLE Algorithm (single time Point; ATE)

1. **Initial estimates.** Fit any predictive model you like to get $\hat{Q}_0(a, w)$ (e.g., GLM, splines, random forests, **Super Learner**). Fit a model for $\hat{g}(w) = \Pr(A = 1 \mid W)$.
2. **Compute the clever covariate.** $\hat{H}_i = \dfrac{A_i}{\hat{g}(W_i)} - \dfrac{1 - A_i}{1 - \hat{g}(W_i)}$.
3. **Target (fluctuation submodel).** Regress Y on $\hat{H}$ with **offset** $\text{logit}(\hat{Q}_0(A, W))$ using logistic loss (for binary Y). This fits a single parameter ϵ:

$$\text{logit}(\hat{Q}_\epsilon(A, W)) \;=\; \text{logit}(\hat{Q}_0(A, W)) + \epsilon \, \hat{H}(A, W).$$

(For continuous Y, use a linear/least-squares fluctuation: $\hat{Q}_\epsilon = \hat{Q}_0 + \epsilon \, \hat{H}$.)

4. **Update the outcome regression.** Set $\hat{Q}^* = \hat{Q}_{\hat{\epsilon}}$. This is the **targeted** prediction function.
5. **Plug in.**

$$\hat{\psi}_{\text{TMLE}} \;=\; \frac{1}{n} \sum_{i=1}^{n} \{\hat{Q}^*(1, W_i) - \hat{Q}^*(0, W_i)\}.$$

6. **Variance/SEs.** Compute the empirical EIF with $\hat{Q}^*, \hat{g}, \hat{\psi}_{\text{TMLE}}$; the sample variance of D^*/n gives standard errors and CIs.

Why this works: The little ϵ-update pushes $\hat{Q}_0$ in the direction that **exactly removes the first-order bias** of your plug-in estimator. If either $\hat{Q}_0$ is good or $\hat{g}$ is good, the EIF mean is driven to 0, giving **double robustness**. When both are good, you get **efficiency**.

A Tiny, Concrete Numeric Example (with Continuous Y to Keep Arithmetic clean)

We'll make Y continuous so the **linear** fluctuation is just one line of algebra. (For binary Y, the idea is identical but uses the logistic link to keep $[0, 1]$ bounds.)

Toy data (8 people):

i	W	A	Y
1	0	1	4
2	0	0	2
3	0	0	1
4	0	0	2
5	1	1	7
6	1	1	6
7	1	1	6
8	1	0	3

- Estimated propensities (by stratum): $\hat{g}(W{=}0) = 0.25$ (1/4 treated), $\hat{g}(W{=}$ = 0.75 (3/4 treated).
- **Initial (misspecified) outcome model** $\hat{Q}_0$: regress Y on A **ignoring** W, giving $\hat{Q}_0(1,\cdot) = \bar{Y}_{A=1} = 5.75$, $\hat{Q}_0(0,\cdot) = \bar{Y}_{A=0} = 2.00$.

Clever covariate H**:** - If $W = 0$: $H = 4$ for treated, $H = -1/0.75 = -1.\overline{3}$ for untreated. - If $W = 1$: $H = 1/0.75 = 1.\overline{3}$ for treated, $H = -1/0.25 = -4$ for untreated.

Residuals $r_i = Y_i - \hat{Q}_0(A_i,\cdot)$**:** - For treated, $r = Y - 5.75$; for untreated, $r = Y - 2.00$.

Linear fluctuation (Gaussian loss):

$$\hat{\epsilon} \;=\; \frac{\sum_i H_i\,\{Y_i - \hat{Q}_0(A_i,\cdot)\}}{\sum_i H_i^2} \;\approx\; -0.172.$$

Update $\hat{Q}_0$ **to** $\hat{Q}^* = \hat{Q}_0 + \hat{\epsilon}\,H$**:** - $\hat{Q}^*(1, W{=}0) = 5.75 + (-0.172)\times 4 \approx 5.06$ - $\hat{Q}^*(0, W{=}0) = 2.00 + (-0.172)\times(-1.\overline{3}) \approx 2.23$ - $\hat{Q}^*(1, W{=}$ = $5.75 + (-0.172)\times 1.\overline{3} \approx 5.52$ - $\hat{Q}^*(0, W{=}$ = $2.00 + (-0.172)\times(-4) \approx 2.69$

Plug-in ATE:

$$\hat{\psi}_{\text{TMLE}} = \frac{1}{2}(5.06{-}2.23) + \frac{1}{2}(5.52{-}2.69) \approx 2.83.$$

What happened?

- The initial (wrong) g-comp ignoring W gave $5.75{-}2.00 = 3.75$ (biased).
- **Targeting** nudged $\hat{Q}$ just enough (using g) to correct that bias *in the EIF direction*, yielding a more plausible marginal contrast. With larger samples (and logistic fluctuation for binary Y), TMLE converges to the truth if either Q or g is correct.

  Teaching note: we used a **linear** fluctuation for easy arithmetic. For **binary** Y you use the **logistic** fluctuation $\text{logit}(\hat{Q}^*) = \text{logit}(\hat{Q}_0) + \hat{\epsilon}\,H$, which keeps predictions in $[0,1]$.

Why TMLE Improves on IPTW and G-computation

- **Versus IPTW:** TMLE avoids **huge weights** at the last step by working through a **bounded** plug-in of Q^*. It still uses g, but primarily to *target* Q rather than to weight raw outcomes; this typically reduces variance and improves stability in finite samples.
- **Versus g-computation:** A naïve plug-in $\frac{1}{n}\sum\{\hat{Q}_0(1,W)-\hat{Q}_0(0,W)\}$ is **biased** if $\hat{Q}_0$ misses important structure. TMLE's **targeting** uses g to eliminate the **first-order bias**—hence **double robustness** (consistent if Q or g is right).
- **Efficiency:** When both Q and g are well learned (e.g., with **Super Learner** and **cross-fitting**), TMLE is **asymptotically efficient** and delivers tight, valid CIs based on the EIF.

Practical TMLE Workflow (R)

Minimal ATE with bounded binary Y using **tmle**:

```
# --- 1. Load Required Libraries ---
# This ensures all necessary packages are ready.
# install.packages(c("tmle", "SuperLearner", "glmnet")) # Run this once if not
installed
# library(tmle)
# library(SuperLearner)

# --- 2. Data Simulation (Placeholder for your actual data) ---
# The error "object 'W' not found" happens because no data exists.
# This section creates a sample dataset so the code can run.
# In your real analysis, you would replace this with code to load your data.
set.seed(456) # for reproducibility
n <- 500 # number of observations

# Create a dataframe of 3 covariates (W)
W <- data.frame(
  W1 = rnorm(n),
  W2 = rnorm(n),
  W3 = rbinom(n, 1, 0.5)
)

# Create a treatment variable (A) that is dependent on the covariates (confounding)
g_true <- plogis(0.75 * W$W1 - 0.5 * W$W2 + 0.7 * W$W3)
A <- rbinom(n, 1, prob = g_true)
```

```r
# Create a binary outcome (Y) that is dependent on treatment and covariates
# The true ATE is approximately 1.0 in this simulation
q_true <- plogis(1.0 * A + 0.6 * W$W1 + 0.4 * W$W2 - 0.5 * W$W3)
Y <- rbinom(n, 1, prob = q_true)


# --- 3. TMLE Analysis ---
# This is the code we corrected previously.

## Super Learner Libraries (A small, fast library for demonstration)
SL.lib <- c("SL.mean", "SL.glm", "SL.glmnet")

# Run the TMLE analysis
fit <- tmle(
  Y = Y,
  A = A,
  W = W,
  family = "binomial",
  Q.SL.library = SL.lib,
  g.SL.library = SL.lib,
  gbound = c(0.01, 0.99)
)


# --- 4. Report Results ---
# Extract and print the key findings
ate_result <- fit$estimates$ATE
psi <- ate_result$psi
se <- sqrt(ate_result$var.psi)
p_value <- ate_result$pvalue
ci_low <- psi - 1.96 * se
ci_high <- psi + 1.96 * se

# Print a clean, formatted summary of the ATE
cat("TMLE Results for the Average Treatment Effect (ATE):\n")
```

```
TMLE Results for the Average Treatment Effect (ATE):
```

```r
cat("  Estimate:", round(psi, 4), "\n")
```

```
  Estimate: 0.1509
```

```r
cat("  Standard Error:", round(se, 4), "\n")
```

```
  Standard Error: 0.0443
```

```r
cat(
  "  95% Confidence Interval: [",
  round(ci_low, 4),
  ", ",
  round(ci_high, 4),
  "]\n"
)
```

```
  95% Confidence Interval: [ 0.064 ,  0.2378 ]
```

```r
cat("  P-value:", round(p_value, 4), "\n")
```

```
  P-value: 7e-04
```

Key points:

- **Super Learner** can flexibly learn both Q and g.
- The package handles the **logistic fluctuation** and EIF-based CIs.
- For small samples or complex learners, prefer **cross-fitting** (e.g., in the **tmle3** / **tlverse** ecosystem) to relax empirical process conditions.

For longitudinal treatments, see **ltmle** (longitudinal TMLE) which handles time-varying A, L, and censoring weights.

Diagnostics and Good Habits

- **Check the clever covariate & EIF:** the empirical mean of the estimated EIF should be very close to 0; the summary typically reports this.

- **Propensity truncation:** if $\hat{g}(W)$ is near 0/1, use truncation bounds (e.g., 0.01, 0.99). Extreme propensities signal **positivity** stress.
- **Cross-fitting:** split the sample to fit Q and g on folds **separate** from where you target and evaluate. This improves robustness with ML.
- **Bounded outcomes:** for binary Y, ensure logistic fluctuation is used so predicted probabilities stay in $[0, 1]$.
- **Sensitivity:** as with IPTW and g-comp, results live or die by **measured confounding** and **overlap**.

How TMLE Relates to IPTW and AIPW

- The **augmented IPTW (AIPW) / one-step** estimator adds the EIF to a preliminary estimator; TMLE instead **targets a plug-in** so the EIF mean is ~0. Under regularity, **AIPW and TMLE are asymptotically equivalent** (both efficient and doubly robust).
- In finite samples, TMLE's **bounded plug-in** can behave better (e.g., respect $[0, 1]$ for probabilities), and the fluctuation step can reduce bias more reliably.

Extensions you'll Likely Use

- **Continuous or multinomial treatments:** generalized clever covariates and link functions.
- **Time-varying treatments (LTMLE):** longitudinal clever covariates (products over time) and sequential targeting.
- **Stochastic / dynamic interventions:** replace static $A = a$ with rules $d(W)$ or stochastic assignments; TMLE extends seamlessly.
- **Survival outcomes:** hazard-based TMLE with censoring handling.

References for Deeper Study

- van der Laan, M. J., & Rubin, D. (2006). *Targeted maximum likelihood learning.*
- van der Laan, M. J., & Rose, S. (2011/2018). *Targeted Learning* (Vols. I & II).
- Gruber, S., & van der Laan, M. J. (2012). *tmle for survival.*
- Schuler, M. S., & Rose, S. (2017). *Targeted methods for causal inference in observational studies.*

TL;DR

TMLE = **g-comp + IPTW + a tiny, principled correction**.

It uses the propensity g to **target** the outcome regression Q so that the **efficient influence function** has mean zero. The result is a **doubly robust**, often **more stable**, and **efficient** estimator of the causal effect—exactly what you want when you care about credible inference with flexible machine learning.

TMLE for the ATE: Rationale, Math, and Proof Sketches (in Plain English)

We'll treat the **average treatment effect (ATE)** with a single binary treatment to keep notation tight. TMLE generalizes to multi-time treatments and other parameters, but the core ideas are the same.

Setup, Target, and Assumptions

- Observed data: $O = (W, A, Y)$ i.i.d. from P_0.
- Treatment $A \in \{0, 1\}$; covariates W; outcome Y (binary or bounded in $[0, 1]$ for this exposition).
- **Target parameter (ATE)**:

$$\psi_0 \;=\; \mathbb{E}_{P_0}[\,Y^1 - Y^0\,] \;=\; \mathbb{E}_{P_0}[\,Q_0(1, W) - Q_0(0, W)\,],$$

where $Q_0(a, w) := \mathbb{E}_{P_0}[Y \mid A{=}a, W{=}w]$ is the **outcome regression**.

- **Nuisance functions**:

$$Q_0(a, w) = \mathbb{E}[Y \mid A = a, W = w], \qquad g_0(w) = \Pr(A = 1 \mid W = w).$$

- **Identification assumptions** (same as g-computation/IPTW): NaN. **Consistency**; 2) **(No unmeasured) exchangeability**: $Y^a \perp A \mid W$; 3) **Positivity**: $0 < g_0(W) < 1$ a.s.

  Plain English. We want a population average contrast of "treat everyone" vs "treat no one." We can express that average in terms of two nuisance functions: how outcomes behave under $A = a$ (the Q function) and how treatment is assigned (the g function).

2) Efficient Influence Function (EIF): What It Is and Why We Care

For the ATE, the (nonparametric) **efficient influence function** is

$$D^*(O; \psi, Q, g) \;=\; H(A, W)\,\{Y - Q(A, W)\} \;+\; (Q(1, W) - Q(0, W)) - \psi,$$

with the **clever covariate**

$$H(A, W) \;=\; \frac{A}{g(W)} \;-\; \frac{1 - A}{1 - g(W)}.$$

Two key properties:

NaN. **Mean zero at truth.** For the true P_0,

$\mathbb{E}_{P_0}[D^*(O; \psi_0, Q_0, g_0)] = 0.$

NaN. **Pathwise derivative.** The EIF is the canonical gradient of ψ at P_0; any regular, asymptotically linear estimator of ψ_0 has influence function with variance $\geq \mathrm{Var}_{P_0}\{D^*(O; \psi_0, Q_0, g_0)\}$. If we construct an estimator whose asymptotic linear representation uses D^*, it is **semiparametrically efficient**.

> **Plain English.** The EIF is the exact direction in which your estimator needs to be "balanced" so that it is unbiased to first order and as precise as possible. If we build an estimator whose random error looks like the average of D^*, we've reached the best possible variance among regular estimators.

3) Rationale of TMLE (Why It Works)

- A **plug-in** estimator $\hat{\psi} = \frac{1}{n}\sum_i \{\hat{Q}(1, W_i) - \hat{Q}(0, W_i)\}$ is natural but biased if $\hat{Q}$ is misspecified.
- **TMLE idea:** start with initial estimates $\hat{Q}_0$ and $\hat{g}$. Then **target** (slightly adjust) $\hat{Q}_0$ along a **one-dimensional submodel** so that the empirical mean of the EIF is (approximately) zero. This removes first-order bias and yields an **asymptotically efficient** plug-in.
- For binary Y, use a **logistic fluctuation** (to keep predictions in $[0, 1]$):

$$\mathrm{logit}\,(\hat{Q}_\epsilon(A, W)) = \mathrm{logit}\,(\hat{Q}_0(A, W)) \; + \; \epsilon\,\hat{H}(A, W),$$

where $\hat{H}(A, W) = \frac{A}{\hat{g}(W)} - \frac{1-A}{1-\hat{g}(W)}$. Estimate ϵ by logistic regression of Y on $\hat{H}$ with **offset** $\mathrm{logit}\{\hat{Q}_0(A, W)\}$. Set $\hat{Q}^* = \hat{Q}_{\hat{\epsilon}}$.

- **Final TMLE:** $\hat{\psi}_{\mathrm{TMLE}} = \dfrac{1}{n}\displaystyle\sum_i \{\hat{Q}^*(1, W_i) - \hat{Q}^*(0, W_i)\}.$

> **Plain English.** We learn an initial outcome model Q and a treatment model g. Then we give Q a tiny, laser-focused nudge in the direction the propensity says is needed so that the key balance condition (mean EIF ≈ 0) holds. Finally, we take the average of the updated potential outcomes. That "nudge" is exactly what makes the plug-in both robust and efficient.

4) Why the Logistic Fluctuation Targets the EIF

Consider the logistic submodel

$$\text{logit}\,(Q_\epsilon(a,w)) = \text{logit}\,(\hat{Q}_0(a,w)) + \epsilon\,\hat{H}(a,w).$$

The **score** at $\epsilon = 0$ for this submodel is

$$\frac{\partial}{\partial\epsilon}\ell(\epsilon)\bigg|_{\epsilon=0} = \frac{1}{n}\sum_{i=1}^{n}\hat{H}(A_i,W_i)\,\{Y_i - \hat{Q}_0(A_i,W_i)\}.$$

Setting this to zero (by logistic regression with offset) makes the **first term** in the empirical mean of the EIF vanish:

$$\frac{1}{n}\sum_i \hat{H}(A_i,W_i)\,\{Y_i - \hat{Q}^*(A_i,W_i)\} \approx 0.$$

The remaining part of the empirical EIF is

$$\frac{1}{n}\sum_i \left(\hat{Q}^*(1,W_i) - \hat{Q}^*(0,W_i) - \hat{\psi}_{\text{TMLE}}\right) = 0$$

by definition of the plug-in $\hat{\psi}_{\text{TMLE}}$. Hence the **empirical EIF mean is ~0** after targeting.

> **Plain English.** The logistic "offset + 1 covariate" regression is chosen so that its score equation is *exactly* the piece of the EIF that involves residuals $Y - \hat{Q}$. When you fit that one parameter, you're solving the EIF equation.

5) Double Robustness: Proof Sketch

Define the **remainder term**

$$\text{Rem}(\hat{Q},\hat{g}) \;=\; \mathbb{E}_{P_0}\Big[\Big(\frac{g_0(W) - \hat{g}(W)}{\hat{g}(W)(1-\hat{g}(W))}\Big)\cdot\{Q_0(A,W) - \hat{Q}(A,W)\}\cdot(2A-\Big].$$

One can show (via standard algebra replacing conditional expectations) that the TMLE (and the one-step/AIPW) admits the expansion

$$\hat{\psi} - \psi_0 = (\mathbb{P}_n - \mathbb{P}_0)\,D^*(O;\psi_0,Q_0,g_0) \;+\; \text{Rem}(\hat{Q},\hat{g}) \;+\; o_p(n^{-1/2}).$$

- The **first term** is mean-zero with variance $\text{Var}\{D^*\}/n$.

- The **remainder** is **second order** in the sense that

$$|\mathrm{Rem}(\hat{Q}, \hat{g})| \;\lesssim\; \|\hat{Q} - Q_0\| \, \|\hat{g} - g_0\|$$

(in an $L_2(P_0)$-type norm, up to positivity bounds). Thus, if **either** $\hat{Q} \to Q_0$ or $\hat{g} \to g_0$ at a reasonable rate (e.g., one of them is consistent and the other is bounded), the product is $o_p(n^{-1/2})$. This yields:

- **Double robustness:** $\hat{\psi}_{\mathrm{TMLE}} \xrightarrow{p} \psi_0$ if **either** Q **or** g is correctly specified/learned.

  Plain English. The only bias left after we balance the EIF is a product of the two errors: how wrong Q is times how wrong g is. If either one is right (or good enough), that product is tiny, so the estimator is still consistent.

6) Asymptotic Normality and Efficiency: Proof Sketch

Under mild regularity (bounded outcomes, positivity, Donsker or **cross-fitting** to avoid entropy issues with ML), we get the **asymptotic linear representation**:

$$\sqrt{n}\,(\hat{\psi}_{\mathrm{TMLE}} - \psi_0) \;=\; \frac{1}{\sqrt{n}} \sum_{i=1}^{n} D^*(O_i; \psi_0, Q_0, g_0) \;+\; o_p(.$$

Hence,

$$\sqrt{n}\,(\hat{\psi}_{\mathrm{TMLE}} - \psi_0) \;\xrightarrow{d}\; \mathcal{N}(0,\; \mathrm{Var}_{P_0}\{D^*(O; \psi_0, Q_0, g_0)\}).$$

Because D^* is the **efficient** influence function, this variance is the **semiparametric efficiency bound**. Thus TMLE is **asymptotically efficient** when both nuisance functions are consistently estimated at adequate rates.

> **Plain English.** After targeting, the estimator behaves like an average of the EIF plus vanishing noise. That gives you normal sampling behavior with the smallest possible asymptotic variance.

7) Relationship to AIPW / One-Step

The AIPW (augmented IPTW, "one-step") estimator is

$$\hat{\psi}_{\mathrm{AIPW}} = \frac{1}{n} \sum_i \left[\frac{A_i (Y_i - \hat{Q}(A_i, W_i))}{\hat{g}(W_i)} - \frac{(1 - A_i)(Y_i - \hat{Q}(A_i, W_i))}{1 - \hat{g}(W_i)} + \hat{Q}(1, W_i) - \hat{Q}(0, W_i) \right].$$

It directly **adds the EIF** (with estimated nuisances) to a preliminary plug-in. Under the same conditions, AIPW and TMLE are **asymptotically equivalent** (same limit and variance). TMLE differs in that it is a **bounded plug-in** estimator (e.g., keeps probabilities in $[0, 1]$) and often has superior finite-sample behavior, especially with ML and when enforcing constraints.

> **Plain English.** AIPW fixes bias by adding a correction term; TMLE fixes bias by *retuning* the outcome model and then plugging in. In large samples they coincide; in small samples TMLE's boundedness and targeting often help.

8) Least Favorable Submodel: Why That One?

TMLE updates Q along a **least favorable submodel** (LFS): a one-parameter path through the model such that the **score** for that parameter equals the EIF component that depends on Q. For binary Y, the logistic LFS has score

$$\frac{\partial}{\partial \epsilon} \operatorname{loglik}(\epsilon)\Big|_{\epsilon=0} = \frac{1}{n} \sum_i \hat{H}(A_i, W_i)\{Y_i - \hat{Q}_0(A_i, W_i)\}.$$

Solving the score equation sets the empirical mean of this piece to zero; the remaining plug-in step sets the rest to zero. This is exactly how TMLE **solves the EIF estimating equation**.

> **Plain English.** We pick the one-dimensional "direction" where a tiny move in Q changes the estimator in the most relevant way (the EIF direction). Then we move just enough to erase the EIF's average. That's the targeting.

9) A Micro Numeric Sanity Check (Binary Y, Tiny sample)

Suppose $\hat{g}(W)$ is known and $\hat{Q}_0$ is a bit off. Compute $\hat{H}_i$, fit logistic regression:

```
glm(Y ~ 0 + H, family = binomial(), offset = qlogis(Q0hat))
```

Extract $\hat{\epsilon}$, form $\hat{Q}^* = \operatorname{expit}(\operatorname{logit}(\hat{Q}_0) + \hat{\epsilon} H)$, then $\hat{\psi} = \frac{1}{n} \sum_i \{\hat{Q}^*(1, W_i) - \hat{Q}^*(0, W_i)\}$. You will find the empirical average of $\hat{H}(A_i, W_i)\{Y_i - \hat{Q}^*(A_i, W_i)\}$ is ~0 (up to numerical tolerance)—i.e., the EIF mean is solved.

> **Plain English.** After the tiny update, the average of the "cleverly weighted residuals" is basically zero. That's the diagnostic that TMLE did what it should.

10) Practical Notes (that Connect back to the proofs)

- **Cross-fitting** (sample splitting) allows highly adaptive ML for Q and g without Donsker conditions and still gives the EIF expansion.
- **Positivity checks** (propensities away from 0/ are crucial for both theory and practice; truncate if necessary (bias–variance trade-off).
- **Variance & CIs:** use the empirical variance of the estimated EIF $D^*(O_i; \hat{\psi}, \hat{Q}^*, \hat{g})$ divided by n.
- **Extensions:** multi-arm A, continuous A, time-to-event, and longitudinal TMLE (LTMLE) all follow the same EIF–targeting blueprint with appropriate clever covariates.

Why This All Matters (Plain English Takeaway)

- The EIF tells you the **gold-standard direction** for removing bias and achieving best precision.

- TMLE is a recipe that **uses your propensity model to surgically correct your outcome model** so that the EIF's average is zero.

- The math guarantees: if either model is right, you're consistent; if both are right, you're **as efficient as possible**—and you stay within natural bounds (e.g., probabilities in $[0, 1]$).

Pointers for Further Reading

- van der Laan & Rose, *Targeted Learning* (Vol. I & II): full derivations of EIFs, least favorable submodels, and large-sample theory.

- Hernán & Robins, *What If*: conceptual foundations and connections to g-computation/IPTW.

- Papers on cross-fitting/debiased ML (Chernozhukov et al.) connect TMLE's EIF expansion to modern ML guarantees.